background arena in which they were themselves embedded. Only relative relationships would matter, a solution much in the spirit of relationists like Leibniz and Mach. Then, as the theory's ingredients—be they strings, branes, loops, or something else discovered in the course of future research—coalesced to produce a familiar, large-scale spacetime (either our real spacetime or hypothetical examples useful for thought experiments), its being a "something" would be recovered, much as in our earlier discussion of general relativity: in an otherwise empty, flat, infinite spacetime (one of the useful hypothetical examples), the water in Newton's spinning bucket would take on a concave shape. The essential point would be that the distinction between spacetime and more tangible material entities would largely evaporate, as they would both emerge from appropriate aggregates of more basic ingredients in a theory that's fundamentally relational, spaceless, and timeless. If this is how it turns out, Leibniz, Newton, Mach, and Einstein could all claim a share of the victory.

Inner and Outer Space

Speculating about the future of science is an entertaining and constructive exercise. It places our current undertakings in a broader context, and emphasizes the overarching goals toward which we are slowly and deliberately working. But when such speculation turns to the future of spacetime itself, it takes on an almost mystical quality: we're considering the fate of the very things that dominate our sense of reality. Again, there is no question that regardless of future discoveries, space and time will continue to frame our individual experience; space and time, as far as everyday life goes, are here to stay. What will continue to change, and likely change drastically, is our understanding of the framework they provide— the arena, that is, of experiential reality. After centuries of thought, we still can only portray space and time as the most familiar of strangers. They unabashedly wend their way through our lives, but adroitly conceal their fundamental makeup from the very perceptions they so fully inform and influence.

Over the last century, we've become intimately acquainted with some previously hidden features of space and time through Einstein's two theories of relativity and through quantum mechanics. The slowing of